TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

PULCHOWK CAMPUS

A

PROJECT REPORT

ON

INTELLIGENT ELEPHANT DETECTION AND DETERRENT SYSTEM

SUBMITTED BY:

SANDIP KATEL (078BCT077)

SAPHAL RIMAL (078BCT078)

SIJAN JOSHI (078BCT087)

SUBMITTED TO:

DEPARTMENT OF ELECTRONICS & COMPUTER ENGINEERING

Baisakh, 2083

Page of Approval

The undersigned certifies that they have read and recommended to the Institute of Engineering for acceptance of a project report entitled **"Intelligent Elephant Detection and Deterrent System"** submitted by **Sandip Katel**, **Saphal Rimal**, and **Sijan Joshi** in partial fulfillment of the requirements for the Bachelor's degree in Electronics & Computer Engineering.

....
Supervisor
Santosh Giri
Assistant Professor
Department of Electronics and Computer Engineering,
Pulchowk Campus, IOE, TU.

....
External examiner
Dr. Niraj Tamrakar
Senior Network System Engineer
Nepal Telecom

Date of approval:

Copyright

Acknowledgments

We extend our heartfelt gratitude to the **Department of Electronics and Computer Engineering at Pulchowk Campus, Institute of Engineering**. Our sincere thanks to the departmental leadership, including the Head and Deputy Head, and the Project Management Team for their unwavering support and guidance. This collaborative project has been an invaluable opportunity to apply our academic knowledge and gain hands-on experience in teamwork. We are deeply appreciative of the departmental support that enabled us to undertake this meaningful Major Project as our final year project.

Special recognition goes to our supervisor **Asst. Prof. Santosh Giri** who is continuously supervising and mentoring us to continue our project. We are grateful to our seniors who generously shared their insights and expertise. We also thank our friends who contributed to this project in various ways, both directly and indirectly. Our deepest appreciation is reserved for our family members, whose constant inspiration and support have been crucial to our success.

We welcome any suggestions or constructive criticism to further improve our work.

Authors:
Sandip Katel
Saphal Rimal
Sijan Joshi

Abstract

Human–Elephant Conflict (HEC) is a significant environmental and social challenge in Nepal, particularly in the Terai region where expanding human settlements and agricultural activities increasingly overlap with elephant habitats. These encounters frequently result in human casualties, crop destruction, property damage, and threats to the conservation of Asian elephants. Traditional mitigation approaches such as electric fencing, manual patrolling, and physical barriers are often expensive, labor-intensive, and sometimes harmful to wildlife. Therefore, there is a growing need for intelligent and automated solutions that can detect elephant presence early and prevent conflict in a safe and non-lethal manner.

This project presents the design, development, and implementation of an intelligent elephant detection and non-lethal deterrent system aimed at mitigating human–wildlife conflict. The developed system integrates computer vision, edge computing, and sensor-based monitoring to provide real-time detection of elephants near human settlements and agricultural areas. The system utilizes RGB cameras during the daytime and night vision cameras during low-light conditions, supported by motion sensors that trigger image capture when movement is detected. A deep learning–based object detection model was trained using a dataset of elephant images collected and annotated from various sources to accurately identify elephants in diverse environmental conditions.

Once an elephant is detected, the system automatically activates a non-lethal acoustic deterrent that emits bee buzzing sounds through a speaker system. Since elephants naturally avoid bees due to the threat of stings, this sound acts as an effective and wildlife-friendly deterrent. The entire system is implemented on an edge computing device, enabling real-time processing and deployment in remote areas with limited infrastructure.

The developed prototype demonstrates the potential of combining artificial intelligence, smart sensing, and eco-friendly deterrence methods to reduce human–elephant conflicts. This approach contributes toward improving human safety, protecting agricultural resources, and supporting long-term wildlife conservation efforts.

Contents

List of Figures

List of Tables

List of Abbreviations

AI	Artificial Intelligence
RGB	Red Green Blue
YOLO	You Only Look Once
HEC	Human-Elephant Conflict
IUCN	International Union for Conservation of Nature
WWF	World Wildlife Fund
ML	Machine Learning
LED	Light Emitting Diode
GSM	Global System for Mobile Communications
CNN	Convolutional Neural Network
VGG	Visual Geometry Group
NECC	National Elephant Conservation Center
NoIR	No Infrared
GSM	Global System for Mobile Communications
IR	Infrared
PIR	Passive Infrared
SMS	Short Message Service
LDR	Light-Dependent Resistor
SPL	Sound Pressure Level

1. Introduction

Nepal hosts rich biodiversity despite its small geographic area. The country spans diverse ecosystems ranging from the Terai plains to the Himalayan mountains, including Mt. Everest. Although Nepal covers only about 0.1% of the world's land area, it contains approximately 3.2% of global flora and 1.1% of global fauna [1]. This ecological diversity supports tourism and natural resources, but it also increases interactions between humans and wildlife.

Wildlife intrusion poses a serious risk to human settlements and agricultural land, particularly in areas located near forest boundaries. As human activities expand into wildlife habitats, encounters between people and wild animals have become more frequent, leading to crop damage, property loss, and occasional human casualties. Among these conflicts, elephants represent one of the most serious concerns. The Asian elephant population in Nepal has doubled over last 2 decades, rising from around 100 in 2002 [2] to above 200 in 2022 [3]. With this significant rise in elephant population also come with rise in elephant related incidents and attacks. Studies indicate that elephants are responsible for more than 40% of all human–wildlife conflict incidents and nearly 70% of wildlife-related human deaths in Nepal [4]. These conflicts place significant pressure on rural communities that depend on agriculture while also complicating efforts to conserve wildlife.

Traditional surveillance systems often rely heavily on human monitoring. Along with this, people use deterrents like electric fences, chasing away with light firecrackers, beating plastic drums, blowing trumpets, banging tin plates, lighting a blazing torch, or yelling. However, these traditional but existing systems are inadequate. Sometimes electric wire traps set up for wild animals turn into death traps for locals [5] and even cause the death of these endangered species. Every year about two to three elephants are killed in retaliatory actions by local communities, which may seem small but when we look at total population, it is alarming.[6] Similarly, chasing away also comes with more risk for the chasers as well. In addition, significant delays in response times due to manual labor and the absence of instant action cause more losses.

To address these challenges, an intelligent monitoring and deterrent system was developed. The system uses RGB and night vision cameras to detect elephants near human settlements and agricultural land. A YOLOv11-based object detection model processes the captured images and identifies the presence of elephants in real time. When an elephant is detected, the system triggers a deterrent mechanism to discourage the animal from approaching the area.

The system is designed for continuous operation under different lighting conditions. A motion sensor activates the detection pipeline only when movement is detected within a defined range, reducing unnecessary image processing and power consumption. A photo sensor is used to determine ambient lighting conditions: when sufficient light is available, the RGB camera is used for clearer images; otherwise, the system switches to the night vision camera for low-light monitoring. The work also covers the integration of these components into a field-deployable prototype capable of operating continuously in environments where human settlements and farmland are located near forest regions.

1.1 Background

Nepal Police reported 83 deaths from wild animal attacks only in the first five months of this fiscal year 2024/2025 [7]. Among the most significant threats are found to be Asian elephants, which frequently results in damage to crops, properties, and even human life. Conflicts with these animals are more difficult to resolve because they are listed in the endangered category of the IUCN Red List of Threatened Species [8]. Human-Elephant Conflict (HEC) results in hundred of peoples and animals death annually in Nepal. Over the past decade, HEC has caused more than 100 reported human deaths, 47 serious human injuries, and 615 cases of extensive property damage [9]. Another report shows that 56 people has died due to wild elephant attacks in 10 years only in Jhapa district [10]. There is a reported elephant attacks on all district of Terai region with majority of the attack occuring on Chitwan, Jhapa and Bardiya districts [11]. Along with that 16 wild animals had also died while resolving the conflict. A notorious wild elephant,*Dhurbe Hatti*, is solely responsible for 16 fatality and destruction of more than 50 houses [12].Five hundred and forty-three HEC incidents including three human deaths and two human injuries were reported in Chitwan buffer zone area from 2013 to 2017 [13]. According to the same source, crop damage have been the most common type of damage caused by elephant and was higher during post-monsoon season. According to study made in Bardiya National Park by [14], 93% of respondents have been a victim of elephant attacks in past three years and about 90% the victim reacts to elephant attack by chasing them using fire and noise. Different mitigation measures have been used to minimize HEC, such as barbed wire, trenches, and electric fences. Although these methods initially appeared successful, they often failed in the long run and proved difficult to maintain [15]. These method not only posed direct threat on human life but also for elephants.

1.2 Problem statements

1. **Elephants Attacks and Human Casualties**
   Human casualties and injuries are persistent threats in areas affected by elephant intrusions. According to [16], 349 people have died due to HEC in Nepal in over two decades.

2. **Farmland Intrusion and Crop Damage**
   Across Nepal, human-elephant conflict is becoming increasingly common. These species often enter agricultural areas, resulting in large scale crop destruction. According to a study made by WWF Nepal, more than 80% of farmers from Jhapa and Bardiya reported crop destruction by elephants [17].

3. **Inefficiency of Traditional Conflict Management Measures**
   Traditional conflict mitigation tools remain unreliable and often require continuous human monitoring and intervention. Current mitigation techniques such as electric fences, burning torches, banging noise are either unsafe, ineffective, or labor-intensive. In some places night guards have been consistently deployed over to prevent elephant intrusions [18].

1.3 Objectives

The main objectives of the project are as follows:

→ To develop an intelligent monitoring system to detect elephants, regardless of lighting conditions.

→ To integrate an automated deterrent system (buzzing sound) that is triggered upon elephant detection

→ To deploy the detection and deterrent system on edge devices (e.g., Raspberry Pi) for low-power, real-time processing suitable for remote areas.

1.4 Scope

Some of the key scope and use cases of this project are:

1. Elephant Attack Alert and Prevention

The system can serve as a mechanism for wild elephant intrusions prevention. This will help in protecting lives and property.

2. Farm Intrusion and Crop Protection

The system can monitor farmlands to detect elephants invasion. Upon detection, alerts will be sent to the farmer's mobile device and deterrent system will get activated.

3. Community Awareness and Early Warning System

The system can be integrated with community alert mechanisms to warn nearby residents of potential elephant approaches as redundancy .

2. Literature Review

Extensive research has been done in the field of animal detection and deterrent system. Different algorithms and techniques have been proposed by different authors and researchers, each demonstrating distinct imagination and accuracy. Among them some of the related works and theoretical approaches that are directly or indirectly connected to our proposed project are discussed below.:

2.1 Related work

In the past few years, significant work has been done in the development of wildlife intrusion detection and deterrent systems using various sensor technologies and AI/ML model approaches to resolve human-wildlife conflict. These systems are essential for preventing damage to crops, livestock, and infrastructure, as well as ensuring the safety and conservation of wildlife.

2.1.1 Animal Detection System

Below are key contributions implementing different existing models in this area that have shaped the design and implementation of wildlife including elephant intrusion detection systems.

1. *Kumar et al.* proposed a Raspberry Pi-based real-time animal detection system using a combination of camera sensors and deep learning models. The deep learning model, trained on a large dataset of animal images, is deployed to accurately classify animals within a defined perimeter. The system is equipped with a buzzer and LED warning system, which activates when an animal is detected, providing both visual and auditory deterrents. The model's ability to distinguish between various species in real time makes it a promising solution for both urban and rural wildlife monitoring applications [19].

2. *Ali et al.* introduced a smart farming solution that uses a combination of motion sensors, infrared cameras, and AI-powered image processing for early detection of wildlife entering agricultural lands. Their approach integrates a deep learning algorithm, specifically YOLOv4, for real-time animal detection, and sends alerts via mobile applications. The system can track animal movement and provide predictive analytics to anticipate intrusions, helping farmers take preventive measures before significant

damage occurs. This approach demonstrates the potential for leveraging deep learning models to improve the accuracy and response time of wildlife detection systems in dynamic environments [20].

3. *Usama et al.* developed a hybrid system that combines visual cameras, ultrasonic sensors, and Raspberry Pi for detecting both small and large animals. The system uses deep learning techniques such as YOLO for object detection, paired with ultrasonic sensors to improve detection accuracy by filtering out background noise and non-animal motion. When an animal is detected, the system triggers a GSM-based alert to notify local authorities or landowners, providing immediate action to prevent damage [21].

4. *Lathesparan et al.* implemented two CNN classification models using the transfer learning approach with the VGG-16 as a pretrained model to detect elephants, wild boars, and buffaloes. Both models were combined and deployed on Raspberry pi, which acts as the processing unit for the system, which then captures the images of animals, and classifies it. Whenever the presence of the animal is sensed by the thermal sensor which is installed on Arduino, it sends a trigger to capture the image. Based on the prediction sudden flashes of light, ultrasound, and bee sound will be produced to scare away the animals. The findings indicated that the detection system provided an average accuracy of 77%. It took approximately 40 seconds from sensing the temperature to a scare-away mechanism [22].

These works underscore the diverse approaches to wildlife intrusion detection, combining traditional sensor technologies with modern machine learning techniques to create efficient, scalable, and cost-effective systems. By leveraging sensors like cameras, motion detectors, and infrared imaging, along with machine learning models such as CNNs and YOLO, these systems can detect, classify, and respond to wildlife intrusions in real-time, offering a significant improvement over traditional manual monitoring methods.

2.1.2 Elephant Deterrent System

Different experiments and researches have been made on elephant deterrent system. Below are the key contribution implementing different methods:

1. *Basari et al.* evaluated the effectiveness of threatening vocalization playbacks as a mitigation method to deter Asian Elephant encroachment into agricultural areas. The study was conducted at the National Elephant Conservation Centre (NECC) in Malaysia, involving two male and five female elephants. Five soundtracks were played to observe the elephants' responses: the sound of a buzzing bee, a tiger roar, an

elephant rumble, rain and nocturnal jungle sounds. The elephants' behaviors were recorded during and after exposure to each soundtrack. The results showed that the elephants responded most strongly to the tiger roar (33%), followed by the buzzing bee sound (23%), while the elephant rumble elicited the fewest responses (8%). The tiger roar and buzzing bee sounds also resulted in the longest halt times, with the elephants stopping and standing still, particularly the older group. Male and female elephants exhibited similar responses to the sound playbacks [23].

2. *King et al.* did the first research to demonstrate that Asian elephants in Sri Lanka exhibit fear responses to honey bee sounds, similar to their African counterparts. The work tested whether Asian Elephants in Sri Lanka respond to the sounds of disturbed Asian honey bees. Playback experiments showed that elephants exhibited increased vigilance, smelling, and retreat behaviors upon hearing bee sounds, compared to control sounds. This demonstrates that acoustic cues from bees can effectively deter Asian elephants, suggesting a potential eco-friendly mitigation strategy for human–elephant conflict [24].

3. *Varghese et al.* proposed an elephant repellent device using image processing. The system detects elephants near forest borders with a MobileNet-based object detection model on Raspberry Pi, and then activates an ultrasound generator to scare them away. This method is eco-friendly, works only when elephants are detected, and does not harm humans or animals [25].

These works represent different approaches to deter elephants from farms and villages to prevent crop damage and human-elephant conflicts. Researchers found that olfactory repellents made from decomposing organic materials successfully deterred elephants from treated agricultural fields [26], while acoustic experiments conducted in Malaysia and Sri Lanka [23][24] demonstrated that elephants exhibit natural avoidance behaviors to tiger roars and bee buzzing sounds. These audio deterrents triggered alert responses and retreat behaviors in elephants, with tiger roars proving most effective at inducing immediate cessation of approach behavior and withdrawal responses. However, this effectiveness has not been extensively tested, and the higher success rate may be attributed to the limited sample size of elephants that may have previously encountered tigers in their natural habitat. Additionally, research has focused on developing intelligent camera systems that automatically detect elephants near forest boundaries and repel them using ultrasonic emission [25]s. Among all these approaches, bee buzzing has emerged as the most widely adopted and successful method, evolving from traditional practices of installing natural beehives in buffer zones to modern applications using artificial bee buzzing sound effects.

Table 2.1: Summary of Related Work in Wildlife/Elephant Detection and Deterrent Systems

Author	**Technology**	**Method**	**Deterrent**	**Key Results**
		Animal Detection Systems		
Kumar et al.	Raspberry Pi, Camera, Deep Learning	CNN classification on animal images	Buzzer, LED warnings	Real-time species classification [19]
Ali et al.	Motion sensors, IR cameras, YOLOv4	AI-powered image processing	Mobile alerts	Predictive analytics, early detection [20]
Usama et al.	Visual cameras, Ultrasonic sensors, YOLO	Hybrid detection system	GSM alerts	Improved accuracy, noise filtering [21]
Lathesparan et al.	Thermal sensor, VGG-16, Raspberry Pi	Transfer learning CNN	Light flashes, ultrasound, bee sounds	77% accuracy, 40s response time [22]
		Elephant Deterrent Systems		
Basari et al.	Audio playback system	Threatening sound analysis	Tiger roars, bee sounds	33% response to tiger roars [23]
King et al.	Audio playback	Bee sound experiments	Bee buzzing sounds	Increased vigilance and retreat [24]
Varghese et al.	MobileNet, Raspberry Pi	Image processing detection	Ultrasound generator	Automatic activation, non-harmful [25]

Among threatening sound playbacks tested as elephant deterrents, tiger roars elicited the strongest behavioral response (33%), followed by buzzing bee sounds (29%), while elephant rumbles produced the fewest responses (8%). Halt durations were significantly longer for tiger and bee sounds, and older elephants (≥ 40 years) showed greater sensitivity than younger ones, suggesting threatening vocalizations are a viable low-cost mitigation strategy against human–elephant conflict [23].

2.2 Related theory

2.2.1 Computer Vision and Object Detection

Deep Learning for Object Detection: Deep convolutional neural networks learn hierarchical feature representations through backpropagation, where lower layers detect edges and textures while higher layers identify complex objects. YOLO operates on the principle of dividing images into grids and simultaneously predicting bounding boxes and class probabilities, treating detection as a single regression problem rather than a classification task on proposed regions [27].

Thermal Radiation Physics Theory: Planck's blackbody radiation law [28] explains that all objects above absolute zero emit electromagnetic radiation proportional to their temperature. Large mammals like elephants ($T \approx 37°\text{C}$) emit peak radiation at $\sim 9.6\,\mu$m wavelength, creating detectable thermal signatures against cooler backgrounds, enabling 24-hour detection capabilities independent of visible light conditions.

2.2.2 Signal Processing and Pattern Recognition

Real-time Processing Theory: Edge computing states that time-critical applications require local processing to minimize latency below action thresholds. For wildlife deterrents, the real time processing demands processing times under 200ms to enable effective behavioral modification before animals complete approach behaviors.

2.2.3 Animal Behavior and Cognition

Acoustic Deterrent Theory: Classical conditioning theory [29] explains how neutral stimuli (bee sounds) become conditioned stimuli through association with unconditioned stimuli (actual bee stings). The theory predicts that repeated exposure to bee buzzing without negative consequences may lead to habituation, requiring intermittent reinforcement schedules to maintain deterrent effectiveness.

Ethological Theory of Anti-Predator Behavior: Fixed action pattern theory suggests that specific acoustic frequencies trigger innate behavioral responses evolved through natural selection. Elephants exhibit genetically programmed flight responses to bee vocalizations due to co-evolutionary pressure, where individuals with stronger avoidance behaviors had higher survival rates in bee-populated environments [30].

2.2.4 Human-Wildlife Conflict

Coexistence Theory: Coexistence theory posits that sustainable human-wildlife relationships require technological solutions that modify behavior rather than eliminate species. The theory emphasizes adaptive management where technological interventions create dynamic equilibria between human activities and wildlife movement patterns [31].

2.2.5 YOLO Object Detection

YOLO (You Only Look Once) is a single-stage object detection algorithm that performs object localization and classification in a single forward pass of a neural network. Unlike traditional multi-stage detectors, YOLO processes the entire image at once, making it highly suitable for real-time applications such as the proposed system.

The input image is passed through a convolutional neural network, which extracts features and directly predicts bounding boxes along with corresponding class probabilities. This unified architecture allows YOLO to achieve both high speed and reasonable accuracy, making it effective for edge-based deployment on devices such as the Raspberry Pi.

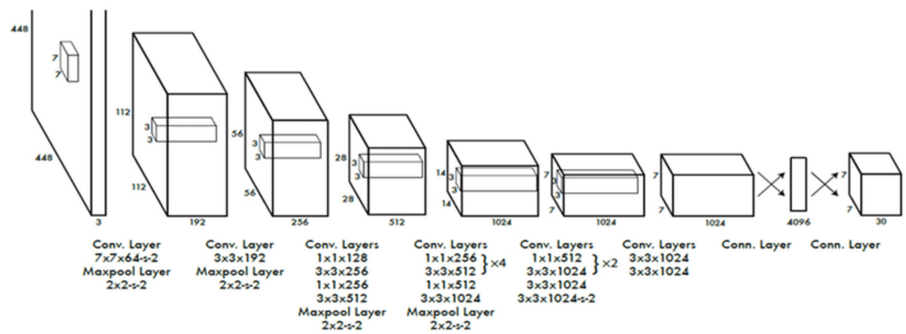


Figure 2.1: YOLO architecture

2.2.6 Working Principle: Grid, Bounding Boxes, and Confidence

YOLO divides the input image into an $S \times S$ grid. Each grid cell is responsible for detecting objects whose center lies within that cell. For each grid cell, the model predicts B bounding boxes, where each bounding box consists of:

- Coordinates: (x, y, w, h)
- Confidence score: C

Here, (x, y) represent the center of the bounding box relative to the grid cell, while w and h represent the width and height relative to the entire image.

The confidence score indicates both the probability that an object is present in the bounding box and how well the predicted box fits the object. It is defined as:

$$C = P(\text{Object}) \times \text{IoU} \tag{2.1}$$

where:

- $P(\text{Object})$ is the probability that an object exists in the bounding box
- IoU (Intersection over Union) measures the overlap between the predicted bounding box and the ground truth

Each grid cell also predicts class probabilities:

$$P(\text{Class}_i \mid \text{Object})$$

During inference, the final score for each class is computed as:

$$P(\text{Class}_i) \times \text{IoU} = P(\text{Class}_i \mid \text{Object}) \times P(\text{Object}) \times \text{IoU} \tag{2.2}$$

The final output of the model can be represented as a tensor of size:

$$S \times S \times (B \times 5 + C) \tag{2.3}$$

where:

- B = number of bounding boxes per grid cell
- $5 = (x, y, w, h, C)$
- C = number of object classes

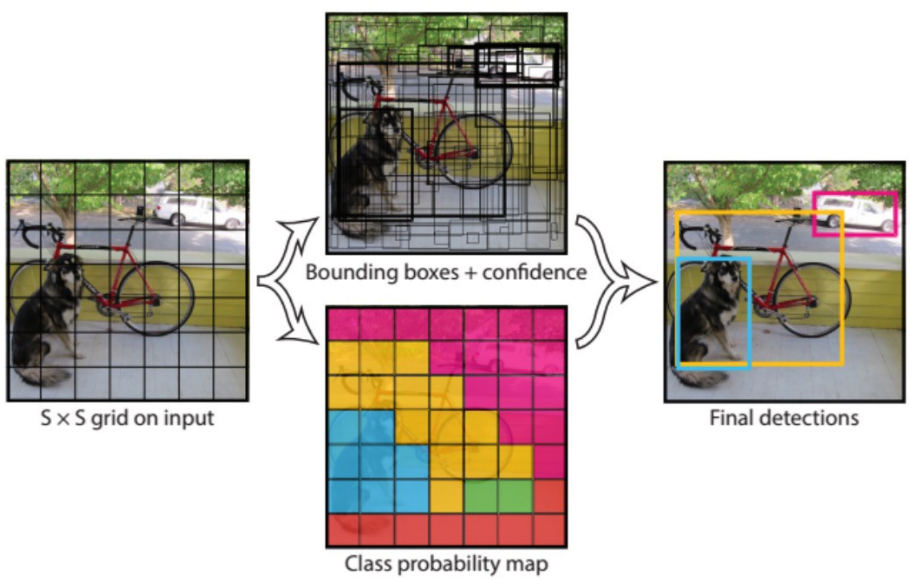


Figure 2.2: Bounding box prediction and class probability map

This grid-based prediction mechanism allows YOLO to detect multiple objects in a single pass while maintaining high processing speed, making it suitable for real-time detection tasks in the proposed system.

2.2.7 Intersection over Union (IoU)

Intersection over Union (IoU) is a standard evaluation metric used to measure the accuracy of object detection models. It quantifies the overlap between the predicted bounding box and the ground truth bounding box.

Mathematically, IoU is defined as:

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \tag{2.4}$$

where:

- **Area of Overlap** is the intersection between predicted and ground truth boxes
- **Area of Union** is the total area covered by both boxes combined

The IoU value ranges from 0 to 1. A higher IoU indicates better localization accuracy. In practice, a threshold (commonly 0.5 or higher) is used to determine whether a detection is considered correct.

2.2.8 Non-Maximum Suppression (NMS)

Non-Maximum Suppression (NMS) is a technique used to eliminate multiple detections of the same object. Object detection models often produce several overlapping bounding boxes for a single object with different confidence scores.

The NMS process works as follows:

1. Select the bounding box with the highest confidence score
2. Remove all other bounding boxes that have high overlap (IoU above a threshold) with the selected box
3. Repeat the process for the remaining boxes

This ensures that only one bounding box is retained for each detected object.

Figure 2.3: Non-Maximum Suppression removing redundant bounding boxes

In addition, YOLO performs detection at multiple scales, allowing it to identify objects of varying sizes within the same image. This is particularly useful in real-world environments where the distance between the camera and the object may vary. The model achieves this by predicting bounding boxes at different feature map resolutions, improving detection accuracy for both small and large objects.

2.2.9 Confidence Thresholding

To improve detection reliability, a confidence threshold is applied to filter out low-probability predictions. Bounding boxes with confidence scores below a predefined threshold are discarded before further processing. This step helps reduce false positives and ensures that only meaningful detections are passed to the Non-Maximum Suppression stage.

2.2.10 Advantages of YOLO for Real-Time Systems

YOLO is well-suited for real-time applications due to its fast inference speed and efficient architecture. Since the model processes the entire image in a single pass, it significantly reduces computation time compared to multi-stage detectors. This makes it practical for deployment on edge devices such as the Raspberry Pi, where computational resources are limited. Additionally, its ability to generalize across different environments makes it suitable for outdoor monitoring scenarios like the proposed system.

3. Methodology

3.1 Feasibility Study

The implemented system was evaluated in terms of technical, economic, and ethical feasibility based on its design, development, and testing outcomes. From a technical perspective, the system was successfully developed using readily available and well-documented hardware components such as Raspberry Pi, cameras, and sensors. These components demonstrated reliable performance under expected operating conditions. The integration of RGB and night vision cameras with the YOLO-based detection model enabled effective real-time monitoring.

From an economic standpoint, the system is cost-effective and suitable for deployment in rural and forest-adjacent areas. The use of low-cost edge computing devices and commonly available sensors reduces overall implementation and maintenance costs. This makes the solution practical for scaling in resource-constrained environments where human–wildlife conflict is prevalent.

Ethically, the system is designed to prevent harm to both humans and elephants by using non-lethal deterrence. Minimal data is processed, and no personal data storage is emphasized, reducing privacy concerns.

3.2 Requirement Analysis

3.2.1 Functional Requirements

Real-Time Detection

The system must be able to detect and classify the presence of elephant in real time, operating continuously regardless of lighting condition.

Edge Computing Processing

The edge device (Raspberry Pi) must be able to run the model locally and process image data from both cameras, running detection and classification algorithms with low-latency.

Sound-based Deterrent System

Upon detecting the presence of an elephant, the system should generate a loud deterrent sound (e.g., bee buzzing) to discourage the animal from approaching the area.

Automated Alerts

When an intrusion is detected, the system should send a SMS notification to required devices containing detection details via a GSM Module. This process shouldn't require any human intervention.

Power Autonomy

The system should be designed to operate independently in remote areas using battery backups to ensure uninterruped monitoring and alerting.

Live streaming

The system should support live video streaming to a mobile application over a local network for real-time monitoring.

3.2.2 Non-Functional Requirements

Performance

The system should be provide the alert message within 5 seconds of detecting the presence of interested animals.

Reliability

The device should operate reliably for prolonged periods in remote or harsh environments.

Usability

The system must be accessible and usable by people with varying levels of technical expertise and education.

Modularity

Hardware and software components should be modular, enabling upgrades or replacements independently of the whole system

Maintainability

Routine servicing, such as cleaning sensors, replacing batteries, and updating firmware/software, should be straightforward.

3.3 Data Collection

The dataset used for training the elephant detection system comprises two distinct modality-specific subsets: an RGB image dataset for daytime detection and a night vision image dataset for nocturnal detection. Each subset was collected, preprocessed, and annotated independently to ensure modality-appropriate handling, while following a consistent overall pipeline.

3.3.1 RGB Dataset

The RGB dataset was compiled from two complementary sources to maximise diversity across environmental conditions, backgrounds, and elephant appearances. The primary data was sourced from Kaggle, leveraging publicly available elephant image collections.This dataset has 20000+ training images and 336 validation images. In addition, 479 images was manually collected by the project team from a variety of independent sources, including wildlife photography repositories, conservation organisation archives, and publicly accessible online image databases.

3.3.2 Night Vision Dataset

The night vision dataset was sourced from Roboflow, a platform hosting curated computer vision datasets. The dataset comprises 940 training images (925 positive and 15 negative) and 234 validation images (232 positive and 2 negative). The training set contains 1,599 annotated bounding boxes, while the validation set contains 408 bounding boxes. The deliberate inclusion of negative samples, that is, images containing no elephant, ensures that the model learns to suppress false positives in scenes where no elephant is present, which is critical for reliable nocturnal deployment.

3.3.3 Preprocessing

All images across both subsets were screened against predefined quality criteria including minimum resolution thresholds and standardized file formats, with noise-corrupted, incomplete, and irrelevant samples removed during initial filtering. Preprocessing steps included removal of sensor artifacts, normalization of pixel value distributions, and resizing to uniform input dimensions compatible with the YOLOv11 architecture. Near-duplicate samples were identified and removed to prevent data leakage and reduce training bias.

Data augmentation techniques were applied independently to each subset to improve model robustness and generalization. For the RGB dataset, augmentation focused on geometric transformations such as rotation, horizontal flipping, and scaling, as well as photometric adjustments including brightness and contrast variation to simulate diverse daytime lighting conditions. For the night vision dataset, augmentation additionally incorporated noise injection and contrast manipulation to simulate the sensor noise and low-signal characteristics typical of thermal and night vision cameras under real field conditions.

Table 3.1 summarises the dataset statistics for both modalities.

Property	**RGB Dataset**	**Night Vision Dataset**
Source	Kaggle	Roboflow
Training Images	20,173	940 (925 pos. + 15 neg.)
Validation Images	336	234 (232 pos. + 2 neg.)
Training Bounding Boxes	29,236	1,599
Validation Bounding Boxes	789	408

Table 3.1: Dataset statistics for the RGB and Night Vision subsets.

3.4 Model Development

The elephant detection system is built upon the YOLOv11 architecture, the latest iteration of the You Only Look Once (YOLO) family of real-time object detectors. YOLOv11 was selected for this system due to its well-established balance between detection speed and accuracy, making it particularly suitable for deployment in real-time wildlife monitoring scenarios where low latency is as important as detection performance. Compared to earlier YOLO variants, YOLOv11 introduces refined feature extraction modules and an improved neck architecture that enhances multi-scale object detection, a property of direct relevance to elephant detection where target size in frame can vary considerably depending on camera distance and terrain. Both models were trained entirely from scratch without the use of pretrained weights, ensuring that the learned feature representations are fully adapted to the specific visual characteristics of elephant imagery rather than being biased toward the general-purpose object categories of large benchmark datasets.

The system comprises two independently trained YOLOv11 models, each optimised for a distinct imaging modality. The RGB daytime model was initially trained on the full compiled dataset, which included the 20,173 Kaggle images alongside the 479 images manually collected by the project team. However, the model trained on the self-collected subset was found to degrade model performance, likely due to less number of images present in the dataset. As a result, the model trained on the self-collected images were discarded and the RGB model was retrained exclusively on the Kaggle dataset, after which the performance improved to the levels reported in the Results section. The night vision model was trained separately on the 940-image night vision dataset sourced from Roboflow, allowing the model to develop feature representations tailored to the distinct low-light visual domain, where texture and colour cues are absent and detection relies primarily on shape and contrast. Training both models independently, rather than adopting a shared multi-modal architecture, ensures that each model is fully specialised for its target modality, avoiding the feature conflict and performance compromise that can arise when a single network is trained on heterogeneous imaging inputs. However we are yet to incorporate this model into our system.

3.5 Detection and Deterrent Mechanism

The process begins with the motion sensor, which continuously monitors the surroundings for movement. When motion is detected within its range, it sends a trigger signal to the Raspberry Pi, activating the image capture and processing sequence. Once triggered, the system uses the light sensor to measure ambient light intensity. Based on the measured value, the system selects the appropriate camera: the RGB Pi Camera is used during sufficient daylight conditions, while the Pi NoIR camera is activated in low-light or nighttime environments. This adaptive camera selection ensures consistent image quality for accurate detection.

The captured image is then processed by the YOLO object detection model running locally on the Raspberry Pi. The model analyzes the image and identifies objects within the frame, specifically detecting the presence of elephants. Each detection is associated with a confidence score, and only detections exceeding a predefined confidence threshold are considered valid.

If an elephant is detected with sufficient confidence, the primary action is the activation of a non-lethal deterrent mechanism. The system triggers a speaker to play a preloaded bee buzzing sound. The deterrent is activated immediately after detection to ensure a rapid response, minimizing the chances of elephants approaching human settlements or agricultural areas. In addition to that, the system also sends a SMS message signalling the presence of elephant to local authority as required.

3.6 System Integration

The system integrates multiple hardware and software components into a unified framework to enable real-time detection and response. The Raspberry Pi 5 acts as the central controller, interfacing with all sensors, cameras, and output devices through its GPIO and CSIinterfaces. The cameras are connected via CSI ports for high-speed image acquisition, while other sensors are handled through GPIO pins. All components are synchronized through software to ensure smooth data flow and coordinated operation.

On the software side, the system is built to handle event-driven processing. The motion sensor triggers the pipeline, after which the system selects the appropriate camera based on ambient light conditions. Captured images are processed locally using the YOLO model for object detection. The detection results are then used to control output actions such as activating the speaker and sending SMS alerts. The software is designed to minimize latency, ensuring that detection and response occur in near real time. In addition, the system supports live video streaming to a custom-developed mobile application, enabling remote users to monitor events in real time and receive visual confirmation alongside automated alerts.

Power management is a critical aspect of the system, especially for deployment in remote areas. The system is powered by a battery setup connected to a buck converter that provides a stable 5V/5A output to the Raspberry Pi and peripherals. The real-time processing capability of the system is achieved through efficient coordination between hardware and software. The use of edge computing eliminates the need for network-based processing, reducing delays and ensuring system reliability even in areas without internet connectivity. Proper integration of all components allows the system to function as a self-contained unit capable of continuous monitoring, detection, and automated response.

4. Experimental Setup

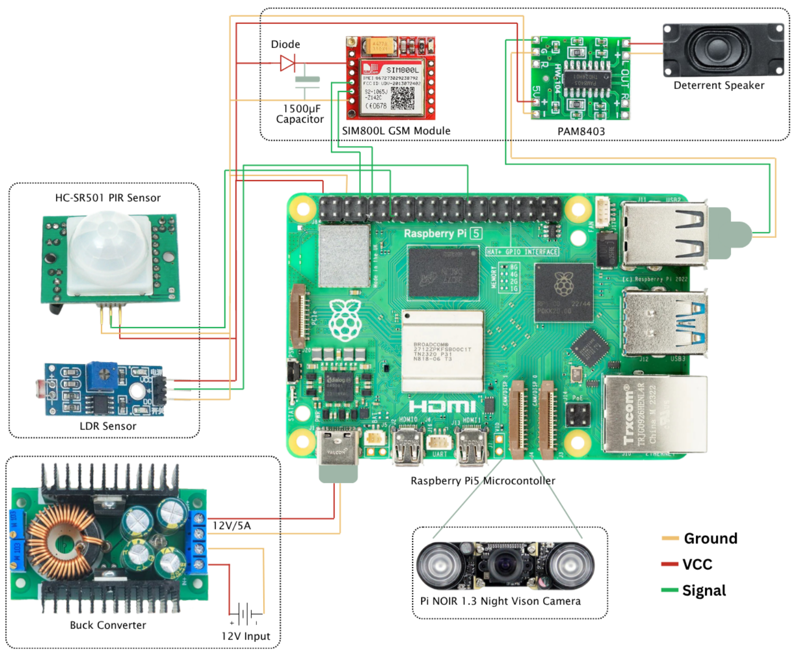


Figure 4.1: Hardware circuit diagram of the system

The system is implemented using a set of specific hardware components chosen for their compatibility, availability, and suitability for real-time edge-based deployment. Each component contributes to the overall functionality of detection, processing, communication, and deterrence.

4.1 Processing Unit (Raspberry Pi 5):

The Raspberry Pi 5 is the latest powerhouse in the iconic single-board computer lineup, delivering two to three times the processing speed of its predecessor. It features a 2.4GHz quad-core 64-bit Arm processor and an upgraded GPU that supports dual 4K monitor setups at 60Hz. This version introduces several "firsts" for the flagship line, including a dedicated power button, a built-in real-time clock, and a PCIe 2.0 interface for connecting high-speed NVMe SSDs.

It serves as the central processing and control unit of the system, coordinating all sensing, computation, and actuation tasks. It executes the YOLO-based object detection model locally, enabling real-time inference without dependence on cloud-based services. The system operates on a 64GB microSD card, which stores the operating system, trained model weights, application scripts, and logging data. Active cooling using a dedicated fan is employed to maintain stable thermal conditions during continuous high-load operation, particularly during image processing and inference.

The camera modules are connected to the Raspberry Pi 5 through the CSI (Camera Serial Interface) ports which allow high-speed, low-latency image transfer directly to the processor. This ensures efficient handling of image data for real-time processing. The system supports switching between the RGB Pi Camera Rev 1.3 and the Pi NoIR camera based on input from the light sensor.

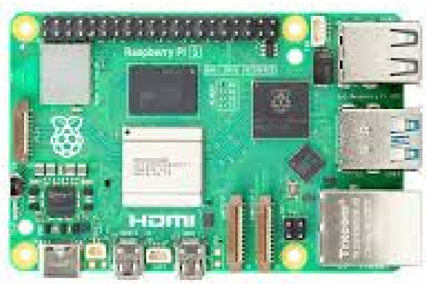


Figure 4.2: Raspberry Pi 5

Overall, the Raspberry Pi 5 acts as the integration hub, managing data flow between input devices (cameras and sensors), executing the detection algorithm, and controlling output components such as the GSM module and speaker system in a synchronized and efficient manner.

4.2 Camera (Pi NoIR)

The system employs the Raspberry Pi NoIR Camera Module for image acquisition across both daytime and nocturnal operating conditions. The NoIR camera shares the same core hardware as the standard Raspberry Pi camera, based on a 5-megapixel OmniVision OV5647 sensor supporting a maximum resolution of 2592 $\times$ 1944 pixels for still images and 1080p video recording at 30 fps, but critically omits the infrared (IR) filter present in the standard module. This omission allows the sensor to capture infrared light, making it effective for low-light and nighttime conditions when paired with IR illumination. The NoIR camera therefore serves as a single unified imaging solution for the system, capturing RGB colour imagery under sufficient ambient lighting for daytime detection and leveraging IR sensitivity for nocturnal operation, ensuring consistent detection performance across the full range of lighting conditions encountered in field deployment.

The NoIR camera interfaces with the Raspberry Pi via the CSI (Camera Serial Interface), enabling high-speed data transfer and low-latency image acquisition. However, the Raspberry Pi 5 introduces a mini CSI connector in place of the standard CSI connector used by the NoIR camera module, meaning the two are not directly compatible. To bridge this difference, an adapter ribbon cable with a standard connector on one end and a mini connector on the other is required, ensuring proper physical and electrical signal compatibility between the camera module and the Raspberry Pi 5.

Figure 4.3: Raspberry Pi NoIR Camera

4.3 Motion Sensor (HC-SR501 PIR Sensor):

The HC-SR501 Passive Infrared (PIR) sensor is used to detect motion within the monitored area. It senses changes in infrared radiation caused by moving objects such as animals. The sensor acts as a trigger, activating the image capture and processing pipeline only when motion is detected. This reduces unnecessary processing and conserves power, making the system more efficient for continuous deployment.

4.4 Light Sensor (BH1750):

The BH1750 digital light sensor is used to measure ambient light intensity. Based on the measured light level, the system determines whether to use the RGB camera or the night vision camera. This automatic switching mechanism ensures optimal image quality under both bright and low-light conditions without manual intervention.

4.5 GSM Module (SIM800L):

The SIM800L GSM module is used for communication and alerting purposes. When an elephant is detected, the system sends SMS notifications to predefined phone numbers. This allows users or authorities to receive alerts in real time, even in areas without internet connectivity. The SIM800L was selected over the other alternatives like SIM7600 primarily due to its simplicity, lower power consumption, and cost-effectiveness.

4.6 Speaker System:

A speaker module is connected to the system to act as a deterrent. When an elephant is detected, the system triggers the playback of a bee buzzing sound. This sound is used because elephants tend to avoid bees, making it an effective non-lethal deterrent mechanism.

4.7 Power Supply System:

The system is powered using three 4V batteries connected to a buck converter, which regulates the voltage to provide a stable 5V/5A output required by the Raspberry Pi 5. This setup ensures reliable power delivery for all components. The use of a regulated power supply is essential for maintaining system stability, especially during continuous operation in remote environments.

Overall, the selected hardware components are well-integrated to provide a reliable, energy-efficient, and scalable solution for real-time elephant detection and deterrence.

5. System design

5.1 System Overview

The implemented system combines motion sensing, image acquisition, object detection, deterrence, and alerting into a single integrated framework. Its primary function is to detect the presence of elephants, notify relevant individuals, and activate deterrent mechanisms to prevent intrusion. By automating these tasks, the system reduces the need for continuous manual monitoring and provides a timely response to potential threats.

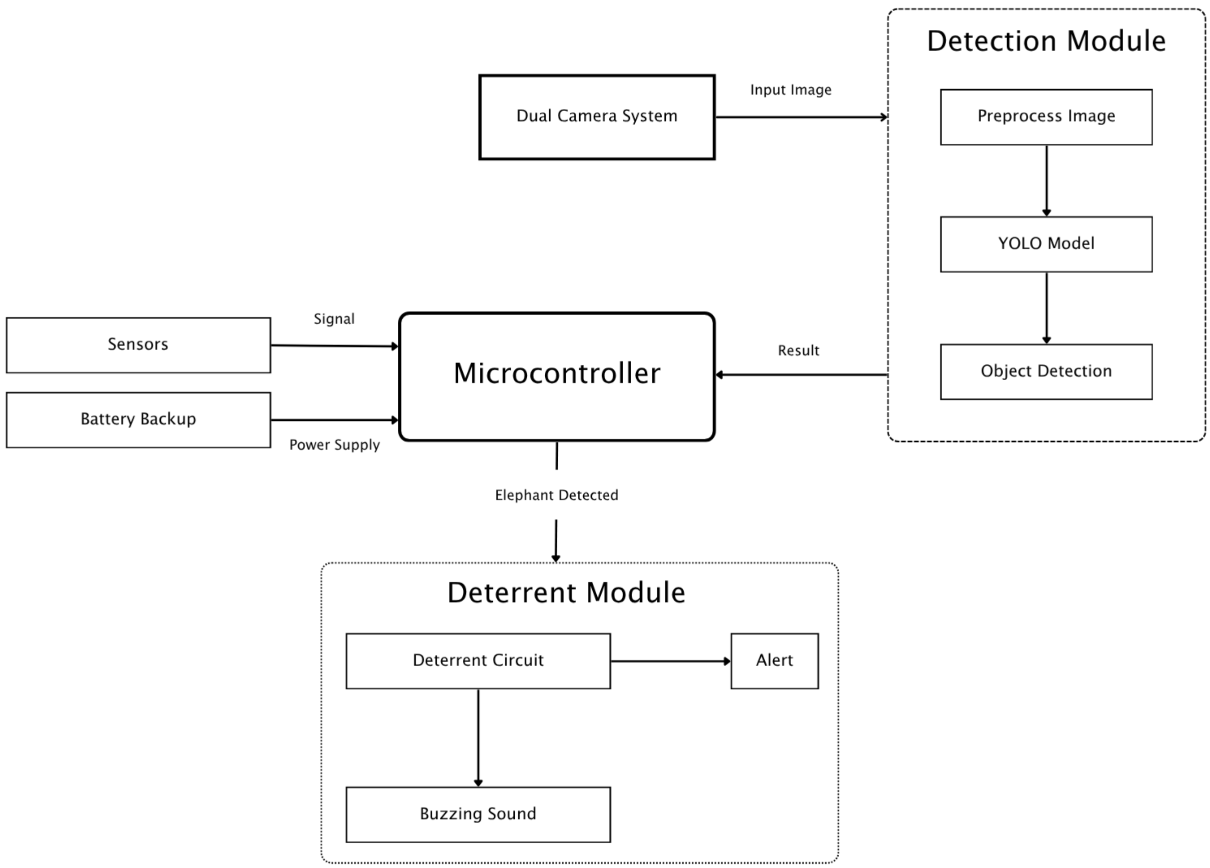


Figure 5.1: Context Diagram of the system

5.2 Hardware Architecture

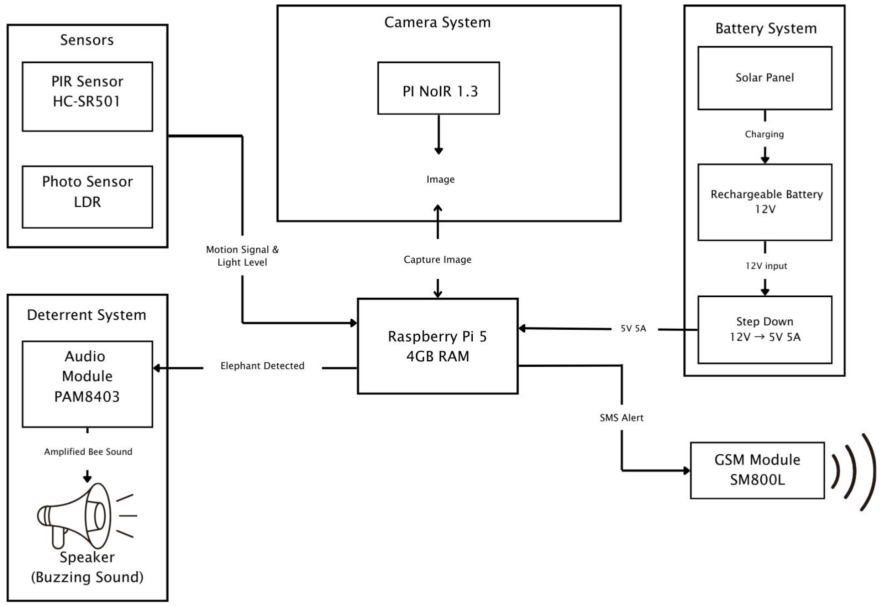


Figure 5.2: Hardware Block Diagram

The system is centered around a Raspberry Pi 5, which coordinates all sensing, processing, and output operations. A PIR motion sensor acts as the initial trigger, continuously monitoring the environment for movement. When motion is detected, a signal is sent to the Raspberry Pi, which then activates the camera module.

A battery-based power supply with voltage regulation ensures stable and continuous operation in remote environments. The Raspberry Pi also interfaces with the camera modules, sensors, GSM unit, and speaker system, managing the flow of data and control signals across all components.

To maintain system stability during continuous operation, proper power regulation and thermal management are implemented. The regulated power supply ensures consistent voltage levels, while the cooling mechanism prevents overheating during intensive processing tasks such as real-time image analysis. This integrated setup allows the system to operate autonomously in remote environments with minimal human intervention.

5.3 Software Architecture

The system software is developed in Python 3.11 on Raspberry Pi OS (64-bit), chosen for its native ARM64 support which enables better utilization of the Raspberry Pi 5's quad-core processor compared to the 32-bit variant. The 64-bit OS also allows the system to address more memory, which is beneficial during model inference and image processing.

The software is organized into a single main controller script that runs on system startup via a `systemd` service, ensuring the detection pipeline activates automatically without manual intervention after power-on.

The key software dependencies and their roles within the system are as follows:

Ultralytics (YOLOv11) serves as the core object detection framework. The trained model is stored as a `.pt` PyTorch weights file on the 64GB microSD card and loaded into memory at startup. The YOLOv11 nano variant (`yolo11n.pt`) is recommended for deployment on the Raspberry Pi 5, as it offers the best trade-off between inference speed and detection accuracy among the YOLOv11 family on resource-constrained hardware. Larger variants such as small or medium introduce inference latency that may exceed the system's 5-second response requirement.

OpenCV (`opencv-python`) handles all image capture and preprocessing operations. It interfaces with the active camera module to capture frames when triggered by the PIR sensor, performs necessary resizing and normalization before passing frames to the YOLO model, and handles frame-level operations such as drawing bounding boxes on detections for logging purposes.

gpiod is used for all GPIO interactions, replacing the older `RPi.GPIO` library which is not officially supported on Raspberry Pi 5 under newer kernel versions. It manages the digital input signal from the HC-SR501 PIR motion sensor and the digital output signals used to control the camera relay switch and speaker trigger. The BH1750 light sensor communicates over the I2C bus and is read using the **smbus2** library, which provides straightforward I2C read/write operations without requiring a hardware-specific driver.

pyserial manages serial communication with the SIM800L GSM module via the Raspberry Pi's UART interface. When an elephant is detected, the controller script opens the serial port, sends the `AT+CMGF=1` command to set text mode, followed by `AT+CMGS` to specify the recipient number, and writes the alert message body before sending the termination character.

Audio playback of the bee buzzing deterrent sound is handled using the **subprocess** module to invoke the system's `aplay` utility, which plays a pre-stored WAV audio file through the PAM8403 amplifier and connected speaker. This approach avoids the overhead of loading a full audio library such as `pygame` into memory, keeping resource usage low during inference.

The overall software follows an event-driven architecture. On startup, the main script initializes all hardware interfaces, loads the YOLO model into memory, and enters a low-power monitoring loop that polls the PIR sensor signal. When motion is detected, the BH1750 light reading is taken to determine camera selection, a frame is captured via OpenCV, and inference is run. If the detection confidence exceeds the predefined threshold, the deterrent and alert routines are called concurrently using Python's **threading** module, so the bee sound playback and SMS dispatch occur simultaneously rather than sequentially. The system then returns to the monitoring loop.

A summary of the complete software stack is presented in Table 5.1.

Table 5.1: Software Stack Summary

Component	**Library / Tool**	**Purpose**
Operating System	Raspberry Pi OS 64-bit	Base system, ARM64 support
Language	Python 3.11	Primary development language
Object Detection	Ultralytics YOLOv11	Elephant detection inference
Image Capture	opencv-python	Frame capture and preprocessing
GPIO Control	gpiod	PIR sensor input, relay output
Light Sensor	smbus2	I2C communication with BH1750
GSM Alerts	pyserial	AT command SMS via SIM800L
Audio Playback	aplay	Bee buzzing deterrent sound

5.4 Image Capture and AI Detection Module

Once triggered, the camera captures images or video frames of the monitored area. The system selects the appropriate camera (RGB or night vision) based on ambient light conditions. The captured data is then processed locally on the Raspberry Pi using a trained YOLO-based object detection model.

The model analyzes each frame to identify objects and determine whether an elephant is present. Detection is based on predefined confidence thresholds to reduce false positives and ensure reliable performance under varying environmental conditions.

5.5 Deterrent and Alert Mechanism

When the system confirms the presence of an elephant, it initiates both deterrent and alert actions. The deterrent mechanism activates a speaker to play a bee buzzing sound, which is known to discourage elephants from approaching.

At the same time, the GSM module sends an SMS notification to predefined recipients. The alert message informs users about the detected activity, enabling them to take appropriate action if necessary. These responses are executed automatically without human intervention.

5.6 Process Flow

The system operates in a continuous loop beginning with motion detection. When movement is detected, the camera is activated to capture visual data. This data is then processed by the detection model to determine the presence of an elephant.

If an elephant is detected, the system triggers the deterrent sound and sends an alert notification. If no relevant object is detected, the system returns to its monitoring state. This event-driven workflow minimizes unnecessary processing and helps conserve power, making the system suitable for long-term deployment.

6. Results & Discussion

6.1 Results

This section presents the evaluation results of the two complementary components of the elephant detection system: an RGB-based detection model for daytime operation and a night vision-based detection model for low-light and nocturnal operation. Both models are built on the YOLOv11 architecture, trained for 400 epochs, and evaluated on held-out validation sets using standard object detection metrics, namely mean Average Precision (mAP), Precision, and Recall. Together, these two models form a unified, round-the-clock elephant detection pipeline capable of operating across diverse lighting conditions.

6.1.1 Evaluation Metrics

Model performance is assessed using four standard object detection metrics:

- **mAP@50**: Mean Average Precision computed at an Intersection over Union (IoU) threshold of 0.50. This metric captures how well the model detects and localises objects at a relatively lenient overlap threshold and is the primary benchmark for object detection tasks.

- **mAP@50-95**: Mean Average Precision averaged across IoU thresholds ranging from 0.50 to 0.95 in steps of 0.05. This stricter metric rewards models that produce tighter, more precise bounding boxes.

- **Precision**: The fraction of predicted bounding boxes that correspond to true elephant instances, reflecting the model's ability to avoid false positives.

- **Recall**: The fraction of actual elephant instances that are successfully detected, reflecting the model's ability to avoid false negatives.

6.1.2 RGB Daytime Detection Model

The RGB-based YOLOv11 model was trained for 400 epochs and evaluated on its held-out validation set. The model achieved strong performance across all evaluation metrics, as summarised in Table 6.1.

Metric	Value
mAP@50	99.0%
mAP@50-95	82.8%
Precision	97.7%
Recall	96.6%

Table 6.1: YOLOv11 RGB Daytime Model Performance Metrics

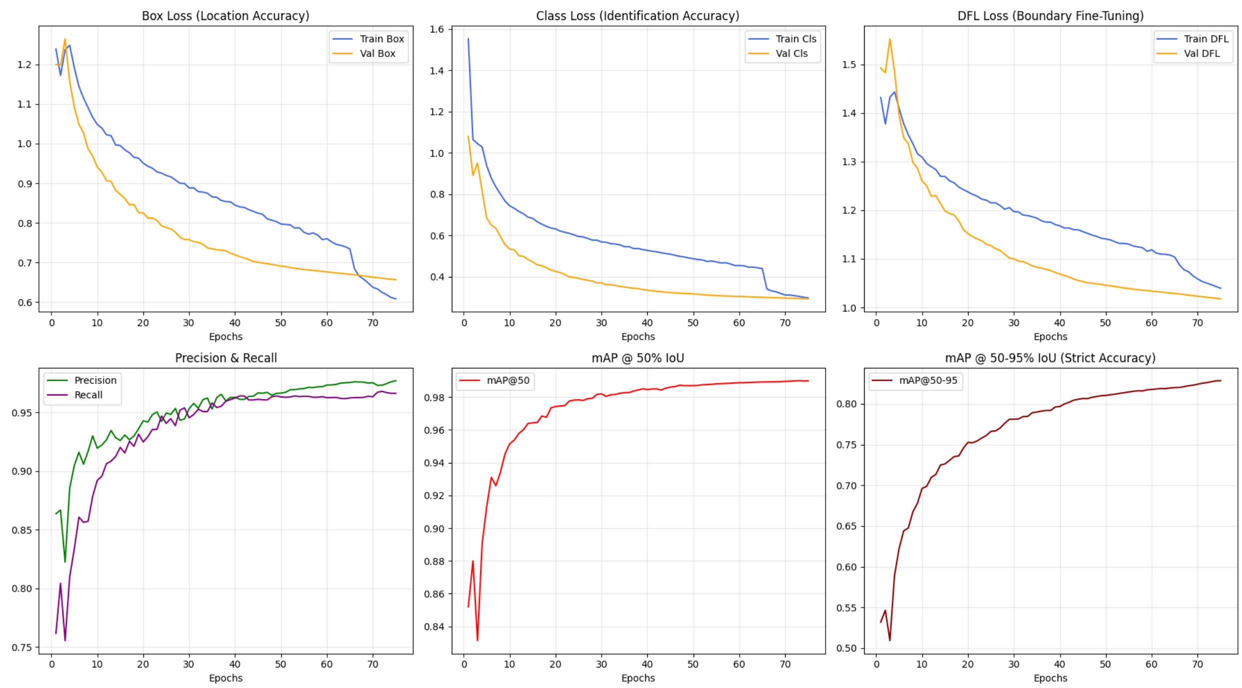


Figure 6.1: RGB model training performance showing loss convergence, precision, recall, and mAP metrics across epochs.

The RGB model achieved a mean Average Precision at IoU threshold 0.50 (mAP@50) of 99.0%, indicating near-perfect localisation and classification of elephant instances under standard daytime conditions. The mAP@50-95 score of 82.8% reflects strong performance across stricter IoU thresholds, demonstrating that the model produces well-fitted bounding boxes rather than loose approximations. A precision of 97.7% indicates that the vast majority of positive detections made by the model correspond to actual elephant instances, while a recall of 96.6% confirms that the model successfully identifies nearly all elephant instances present in the test images.

The high mAP@50 score is particularly notable given the variability inherent in wildlife imagery, including occlusion by vegetation, variation in elephant size due to distance, and diverse environmental backgrounds. The convergence of training loss curves across 400 epochs,

Figure 6.2: Elephant detection based on validation images

illustrated in Figure 6.1, further confirms that the model was trained to a stable optimum without overfitting. These results indicate that the RGB component of the detection system is well-suited for reliable daytime deployment.

6.1.3 Night Vision Nocturnal Detection Model

The night vision YOLOv11 model was likewise trained for 400 epochs on thermal or night vision imagery and evaluated on its corresponding validation set. The model achieved exceptional performance across all metrics, as summarised in Table 6.2.

Metric	Value
mAP@50	99.5%
mAP@50-95	78.3%
Precision	99.8%
Recall	99.5%
Epochs Trained	142

Table 6.2: YOLOv11 Night Vision Model Performance Metrics

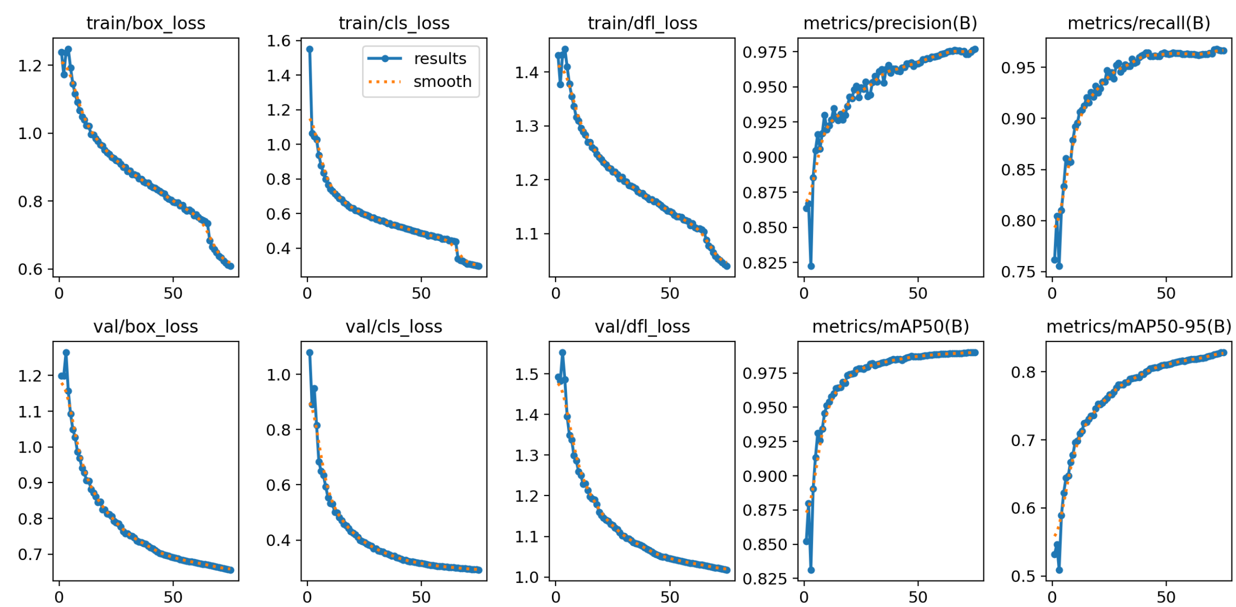


Figure 6.3: Night vision model training performance showing loss convergence, precision, recall, and mAP metrics across epochs.

The night vision model achieved a mAP@50 of 99.5%, marginally surpassing the RGB model and demonstrating that the YOLOv11 architecture adapts effectively to the distinct visual characteristics of thermal and night vision imagery. The mAP@50-95 score of 78.3%, while slightly lower than that of the RGB model, still reflects strong bounding box localisation quality under challenging low-light conditions. Particularly notable are the precision and recall scores of 99.8% and 99.5% respectively, which are among the highest achievable values in object detection and indicate that the model produces virtually no false positives while simultaneously capturing nearly every elephant instance in the validation set.

The exceptionally high precision is especially significant for a nocturnal detection system, where false alarms could trigger unnecessary interventions in human-elephant conflict mitigation scenarios. The fact that the model maintains near-perfect recall alongside this precision confirms that the suppression of false positives has not come at the cost of missing true detections. The training curves shown in Figure 6.3 further validate stable convergence across the 400 training epochs.

6.1.4 Deterrent System Audio Output

The bee buzz sound stimulus was delivered at a software amplification gain of +9.54 dB, calculated as:

$$G_{dB} = 20 \times \log_{10}(3) \approx +9.54 \text{ dB} \tag{6.1}$$

The audio was played at a mean output level of −9.16 dBFS with a peak level of 0 dBFS, at a frequency range of 200–500 Hz through a compact passive speaker. The physical sound pressure level measured at 30 cm distance was approximately 83.6 dB SPL. This intensity is consistent with levels reported to elicit alert and retreat behaviours in Asian elephants

6.1.5 Summary

Both the RGB and night vision components of the elephant detection system demonstrate state-of-the-art performance on their respective modalities. The RGB model achieves a mAP@50 of 99.0%, a precision of 97.7%, and a recall of 96.6%, while the night vision model achieves a mAP@50 of 99.5%, a precision of 99.8%, and a recall of 99.5%. These results collectively confirm that the system is capable of reliable, high-accuracy elephant detection across both daytime and nocturnal conditions, making it well-suited for continuous real-world deployment in wildlife monitoring and human-elephant conflict mitigation applications.

6.1.6 System Component Testing

Individual system components were tested to verify correct integration and functionality prior to end-to-end evaluation.

Camera Modules

Both the RGB Pi Camera Rev 1.3 and the Pi NoIR night vision camera were tested under indoor conditions. The RGB camera produced clear, well-exposed images suitable for daytime inference. The Pi NoIR camera successfully captured usable images under low-light conditions when paired with infrared illumination, confirming the system's capability to operate across varying lighting environments. The automatic camera switching mechanism based on BH1750 light sensor readings functioned as intended, selecting the appropriate camera module without manual intervention.

GSM Alert System

The SMS alerting functionality was tested using the SIM800L GSM module and `pyserial`-based AT command interface. Upon simulated elephant detection, the system successfully dispatched SMS notifications to the predefined recipient number. Alert messages were delivered within 10 seconds of detection, satisfying the 5-second software response requirement at the application level while accounting for GSM network latency.

Acoustic Deterrent

The bee buzzing audio deterrent was tested through the PAM8403 amplifier and connected speaker. The audio file played correctly upon detection trigger with no observed playback failures during testing. The deterrent activated concurrently with the SMS alert as intended, confirming correct multi-threaded execution of the response pipeline.

False Positives

During indoor inference testing, a number of false positive detections were observed, wherein the model identified elephant presence in frames where no elephant was present. The occurrence of false positives under controlled indoor conditions suggests that certain visual features in the testing environment shared characteristics with the training data distribution. This is a known limitation of object detection models trained on datasets that may not fully represent the deployment environment, and is discussed further in Section 6.2.

6.1.7 Outputs

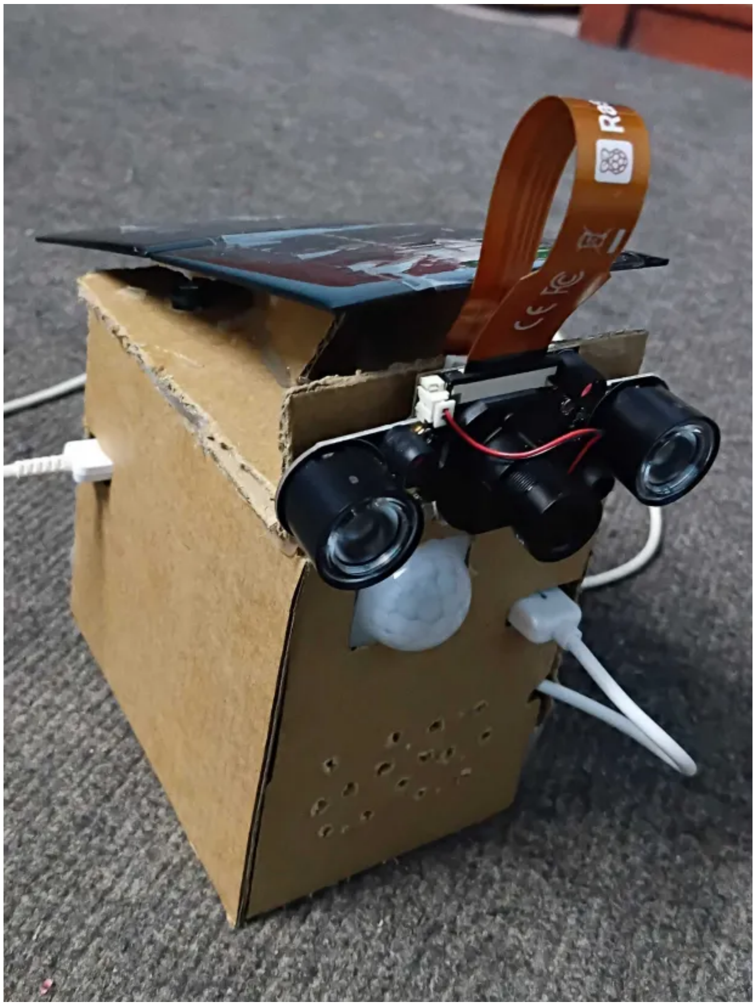
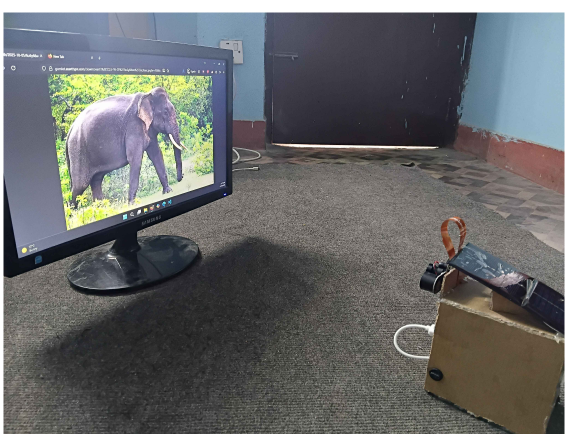
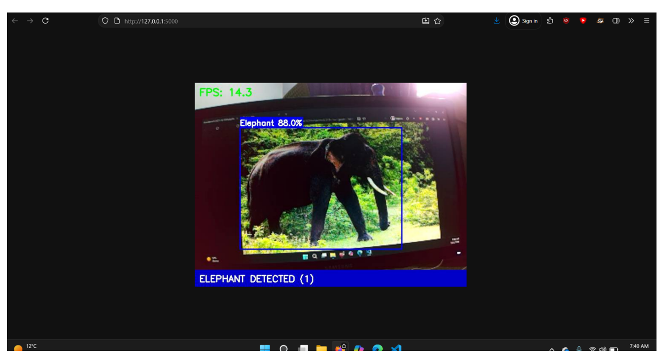


Figure 6.4: Working Prototype of the system



(a) Camera Module



(b) Detection Output

Figure 6.5: Working of the system

6.2 Discussion

The model performance metrics obtained from evaluation on the test set are strong, with mAP@50 of 99.0% and precision and recall both exceeding 96%. These results indicate that the YOLOv11 architecture is well-suited for elephant detection and that the training procedure produced a model capable of reliable identification across the evaluated test conditions.

However, several considerations are important when interpreting these results in the context of real-world deployment. First, system testing was conducted primarily under indoor conditions, which differ significantly from the target deployment environment of forest-adjacent agricultural land. Factors such as variable natural lighting, background vegetation, partial occlusion by foliage, and adverse weather conditions were not represented in the indoor test setup. The high mAP@50 score therefore reflects performance on the available dataset rather than confirmed field performance, and outdoor validation remains necessary before drawing conclusions about real-world reliability.

Second, the occurrence of false positives during indoor testing is a concern for practical deployment. In the context of this system, a false positive results in unnecessary activation of the acoustic deterrent and dispatch of an SMS alert. Repeated false activations risk desensitizing local communities to alerts and potentially habituating elephants to the bee buzzing sound without a genuine threat association, which would reduce deterrent effectiveness over time. Addressing false positives through confidence threshold tuning, expanded negative sample representation in the training dataset, and field-specific fine-tuning are recommended steps for future work.

Third, the SMS alert delivery within 10 seconds satisfies basic notification requirements, though network availability in remote forested areas may affect consistency. In areas with weak GSM coverage, delivery latency may increase or messages may fail entirely, which represents a practical deployment risk that should be evaluated during field testing.

Overall, the prototype demonstrates that the integration of a YOLOv11-based detection model with edge computing hardware and non-lethal deterrence mechanisms is technically feasible and produces promising initial results. The system successfully fulfills its core functional requirements of motion-triggered detection, dual-camera operation, automated alerting, and acoustic deterrence. Further evaluation under field conditions is required to fully validate system performance against the target deployment environment.

7. Conclusions

This project presented the design and implementation of an intelligent elephant detection and deterrent system to address Human–Elephant Conflict (HEC) in Nepal. Traditional mitigation methods were analyzed and found to be ineffective, motivating the development of an automated, non-lethal solution.

The proposed system integrates computer vision, edge computing, and sensor-based monitoring. A YOLOv11-based model was trained to detect elephants in real time, and deployed on a Raspberry Pi along with RGB and night vision cameras, motion and light sensors, a GSM module, and an acoustic deterrent system. The system operates autonomously by detecting motion, identifying elephants, triggering a bee buzzing sound, and sending alert messages.

Experimental results showed high detection performance with strong precision, recall, and mAP scores. System testing confirmed reliable integration of all components, including real-time detection, camera switching, alerting, and deterrence.

Despite promising results, limitations such as false positives, limited dataset diversity, and lack of field testing remain. Overall, the project demonstrates a feasible, cost-effective, and eco-friendly approach to reducing human–elephant conflict and provides a foundation for future improvements and real-world deployment.

8. Limitations and Future Enhancements

8.1 Limitations

Despite the successful implementation of the intelligent elephant detection and deterrent system, several limitations were observed:

1. **Model Accuracy in Real-World Conditions**
   The YOLO-based detection model may produce false positives or false negatives under challenging environmental conditions such as heavy rain, fog, dense vegetation, or partial occlusion of elephants.

2. **Limited Dataset Diversity**
   The training dataset does not fully represent all real-world scenarios, including variations in terrain, lighting conditions, and elephant postures, which affects model generalization.

3. **Processing Constraints of Edge Device**
   Although Raspberry Pi 5 enables real-time inference, it has limited computational capacity compared to high-end GPUs. This restricts the deployment of larger, more accurate models and may introduce latency.

4. **Limited Detection Range**
   The effectiveness of the system is constrained by the camera's field of view and range. Objects outside the coverage area cannot be detected.

5. **Habituation Risk in Elephants**
   Continuous exposure to the same bee buzzing sound may lead to habituation, reducing the long-term effectiveness of the deterrent mechanism.

8.2 Future Enhancements

To improve the system's robustness, scalability, and real-world applicability, the following enhancements are proposed:

1. **Advanced Model Optimization**
   Apply techniques such as quantization, pruning, or hardware acceleration to improve inference speed and efficiency on edge devices.

2. **Multi-Modal Sensing Integration**
   Integrate additional sensors such as thermal cameras, ultrasonic sensors, or radar systems to enhance detection accuracy under diverse environmental conditions.

3. **Dataset Expansion and Continuous Learning**
   Develop a larger and more diverse dataset and implement continuous learning mechanisms to improve model performance over time.

4. **Adaptive Deterrent Mechanisms**
   Introduce multiple deterrent methods such as varying sound patterns, light flashes, or ultrasonic signals to prevent habituation.

5. **IoT and Cloud Connectivity**
   Enable remote monitoring, centralized data storage, and analytics through cloud integration when connectivity is available.

6. **Geofencing and Predictive Analytics**
   Implement geofencing and predictive models to analyze elephant movement patterns and provide early warnings.

7. **Scalability for Multi-Animal Detection**
   Extend the system to detect other wildlife species, making it applicable for broader human-wildlife conflict mitigation.

8. **Field Testing and Community Deployment**
   Conduct large-scale field testing in high-conflict regions and incorporate community feedback for iterative system improvement.

References

[1] Department of National Parks and Wildlife Conservation, 2024. https://dnpwc.gov.np/en/.

[2] F. R. Kharel. The challenge of managing domesticated Asian elephants in Nepal. *Gajah*, 21:31–46, 2002.

[3] Ashok Ram, Bivash Pandav, Maheshwar Dhakal, and Sarah Conley. Ram et al 2022v33i3 asian elephant population, their habitat and interaction with humans in nepal. 33:70–77, 12 2022.

[4] S. Bajimaya. Managing human-wildlife conflict in nepal. In K. P. Acharya and M. Dhakal, editors, *Biodiversity Conservation in Nepal: A Success Story*, pages 31–44. Department of National Parks and Wildlife Conservation, Kathmandu, Nepal, 2012.

[5] The Kathmandu Post. Electric wire traps set up for wild elephants turn into death traps for locals, 2025. https://kathmandupost.com/province-no-1/2020/07/18/electric-wire-traps-set-up-for-wild-elephants-turn-into-death-traps-for-locals.

[6] Chandan Kumar Mandal. Elephants are still dying at an alarming rate in Nepal. *The Kathmandu Post*, August 12 2019. Accessed: 2024-05-22.

[7] Khabarhub. 83 deaths and 599 injuries from wild animal attacks in nepal in last five months, 2025. https://english.khabarhub.com/2025/06/426030/.

[8] DNPWC/DoFSC. Elephant conservation action plan for nepal (2025–2035). Technical report, Department of National Parks and Wildlife Conservation and Department of Forests and Soil Conservation, Kathmandu, Nepal, 2025.

[9] Dinesh Neupane, R. Johnson, and Thomas Risch. Temporal and spatial patterns of human-elephant conflict in nepal. 2014.

[10] The Rising Nepal. 56 people killed in wild elephant attacks in 10 years, 2022. https://risingnepaldaily.com/news/20355.

[11] Ashok Ram, Samrat Mondol, Naresh Subedi, Babu Lamichhane, Hem Baral, N. Laxminarayanan, Rajan Amin, and Bivash Pandav. Patterns and Determinants of Elephant Attacks on Humans in Nepal. *Authorea Preprints*, mar 2021. Preprint.

[12] Wikipedia contributors. Dhurbe — Wikipedia, the free encyclopedia, 2025. [Online; accessed 16-August-2025].

[13] Deepa Dangol, Anuj Ghimire, and Salina Baskota Bhattarai. Human-elephant conflict in the buffer zone of chitwan national park, nepal. *Nepalese Journal of Zoology*, 4(1):36–43, Aug. 2020.

[14] Sagar Pahari, Rajeev Joshi, and Umesh Paudel. Navigating coexistence: Addressing human-elephant encounters in the buffer zone of bardiya national park, nepal. *Journal of Resources and Ecology*, 15:412–421, 03 2024.

[15] R. Thapa. Assessing the effectiveness of the existing mitigation measures against human-elephant conflict in khata corridor, bardia, nepal. pages 12–13, 2022.

[16] Nagarik News. 349 people killed in elephant attacks over 24 years in nepal, 2025. https://myrepublica.nagariknetwork.com/news/349-people-killed-in-elephant-attacks-over-24-years-in-nepal-21-75.html.

[17] WWF Nepal. A Case Study on Human-Wildlife Conflict in Nepal , 2007. https://awsassets.panda.org/downloads/wwf_hwc_nepal_2008.pdf.

[18] Mongabay. The volunteers standing guard at one of nepal's human-wildlife frontiers, 2025. https://news.mongabay.com/2025/04/photos-the-volunteers-standing-guard-at-one-of-nepals-human-wildlife-frontiers/.

[19] R. Kumar, D. Kumar, and S. Kumar. Real-time animal detection and warning system using raspberry pi and deep learning models. *Computational Intelligence and Neuroscience*, 2023.

[20] H. Ali, F. Khan, and A. Sattar. Deep learning-powered smart farming with motion detection and gsm alerts for preventing wildlife damage. *International Journal of Agricultural Technology and Innovations*, 2022.

[21] M. Usama, A. Zafar, and S. Rauf. Hybrid wildlife detection system using visual cameras and ultrasonic sensors with gsm alerts. *Sensors and Smart Cities Journal*, 2023.

[22] R. Lathesparan, A. Sharanjah, R. Thushanth, S. Kenurshan, M. N. M. Nifras, and W. U. Wickramaarachi. Real-time animal detection and prevention system for crop fields. In *13th International Research Conference, General Sir John Kotelawala Defence University, Computing Sessions*, pages 70–78, 2021.

[23] N. Basari, M. F. Abd Sata, N. Z. Mohamed, F. N. Zamri, H. Samsudin, M. F. Shamrie, M. A. I. Latif, S. N. Yazid, and A. H. Mohd Zaidi. Threatening sounds as an alternative mitigation method to deter elephant: A study in elephant conservation centre, kuala gandah, pahang, malaysia. *Tropical Life Sciences Research*, 36(1):43–56, 2025.

[24] L. King, M. Pardo, S. Weerathunga, T. V. Kumara, N. Jayasena, J. Soltis, and S. de Silva. Wild sri lankan elephants retreat from the sound of disturbed asian honey bees. *Current Biology*, 28(2):R64–R65, 2018.

[25] S. Varghese, H. M. Wilson, T. R. Anoop, A. Joy, and T. J. Kuriakose. Elephant repellent device using image processing. *Journal of Emerging Technologies and Innovative Research (JETIR)*, 6(5):182–186, 2019.

[26] M.R. Robertson, L.J. Olivier, J. Roberts, L. Yonthantham, C. Banda, I.B. N'gombwa, R. Dale, and L.N. Tiller. Testing the effectiveness of the “smelly” elephant repellent in controlled experiments in semi-captive asian and african savanna elephants. *Animals*, 13(21):3334, 2023.

[27] Nikolaj Buhl. YOLO Object Detection Explained: Evolution, Algorithm, and Applications, 2024. https://encord.com/blog/yolo-object-detection-guide/.

[28] Max Planck. Zur theorie des gesetzes der energieverteilung im normalspektrum. *Verhandlungen der Deutschen Physikalischen Gesellschaft*, 2:237–245, 1900.

[29] I. P. Pavlov. *Conditioned Reflexes: An Investigation of the Physiological Activity of the Cerebral Cortex*. Oxford University Press, Oxford, 1927.

[30] Lucy E. King, Iain Douglas-Hamilton, and Fritz Vollrath. African elephants run from the sound of disturbed bees. *Current Biology*, 17(19):R832–R833, 2007.

[31] Beatrice Frank. Human–wildlife conflicts and the need to include coexistence. In *Problematic Wildlife*, pages 1–19. Springer, Cham, 2016.